

COMPILING SHORTEST PATHS: FROM RATIONAL PUSHDOWN NORMALIZATION TO FINITE CUBICAL GEODESIC COMPILERS

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ABSTRACT. We develop a method of *cubical geodesic compilation* for positive-cost finite-phase periodic action systems. The method arose from the shortest-completion geometry of a completion-aware 2:1 deterministic pushdown automaton and its coprime $p:q$ generalization: the pushdown dynamics lead to an arithmetic residual, minimum completion becomes a directed shortest-path problem, and families of optimal completions expose both optimal branching and higher-order compatibility among commuting completion actions. We isolate this mechanism from its pushdown origin and formulate it at the level of periodic shortest-path systems.

From finite arithmetic data, the resulting compiler recovers, at every target-reachable state, exact shortest distance, all optimal first moves, shortest-prefix membership for arbitrary finite executable words, the complete local optimal-compatibility complex, canonical shortest routes, and an implicit exact representation of all shortest paths. The arithmetic distance layer belongs to the classical theory of periodic shortest paths; cubical geodesic compilation augments it with exact higher-order optimal-compatibility information.

We then develop a structure theory for the information encoded by the compiler. For every prescribed order $k \geq 1$, there exists a completion-total, one-phase, unit-cost system of globally commuting translation actions with two adjacent states having the same shortest distance and identical optimal-compatibility data through order $k - 1$, while their truncations through order k differ by exactly one k -face. Consequently, for every fixed h , exact distance together with the order- h optimal-compatibility truncation does not universally determine the full local optimal-compatibility complex.

Independent products amplify the separation. For every $k, m \geq 1$, a single common order- $(k - 1)$ signature supports 2^m distinct order- k signatures obtained by independently toggling m distinguished faces. Thus higher-order optimal compatibility is finitely compilable, while at every prescribed order it can contain information not determined by all preceding compatibility levels.

2020 *Mathematics Subject Classification.* Primary 68R10; Secondary 05C85, 68Q25, 68Q45, 90C35.

Key words and phrases. periodic shortest paths; cubical geodesic compilation; deterministic pushdown normalization; Presburger arithmetic; optimal-compatibility complexes; affine semigroups.

1. INTRODUCTION

Shortest-path algorithms are usually organized around metric information: the value of an optimal route, one selected shortest path, or the set of optimal next moves. Structured infinite systems can carry a richer form of local optimality. Several actions may be individually compatible with optimal behavior while failing to occur jointly on any shortest path. Conversely, structurally commuting actions may generate higher-dimensional families of shortest prefixes.

This leads to two complementary questions. First, how much of the optimal behavior of an infinite structured shortest-path system can be represented by finite preprocessing? Second, if one records increasingly high-order compatibility among optimal actions, does this hierarchy eventually collapse to lower-order information?

We approach these questions through a method we call *cubical geodesic compilation*.

1.1. Motivation. The starting point is the completion-aware real-time deterministic pushdown automaton in Figure 1, developed for the weighted balance language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Unlike a recognition-only counter implementation, this machine was organized so that unresolved balance is represented jointly by finite control and the stack and can be materialized, at end of input, as a canonical shortest completion. This completion behavior is the concrete motivation for the shortest-path viewpoint developed in the present paper.

Ignoring the bottom marker, let $S \in \{0,1\}^*$ denote the logical stack of the 2:1 core and define

$$D(S) = 2N_0(S) - N_1(S), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

For an input prefix w , the scalar residual is

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

and the machine carries the distributed invariant

$$\boxed{\rho_{2,1}(w) = D(S) + \Phi(q_{\text{state}})}.$$

Thus the machine-specific stack representation exposes a representation-independent integer quantity. Passing to arbitrary coprime positive integers p, q gives

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\}, \quad \boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w)}.$$

Reading a 1 changes the residual by $+q$, while reading a 0 changes it by $-p$. Hence residual completion is already a translation shortest-path problem on the integer line:

$$r \xrightarrow{1} r + q, \quad r \xrightarrow{0} r - p.$$

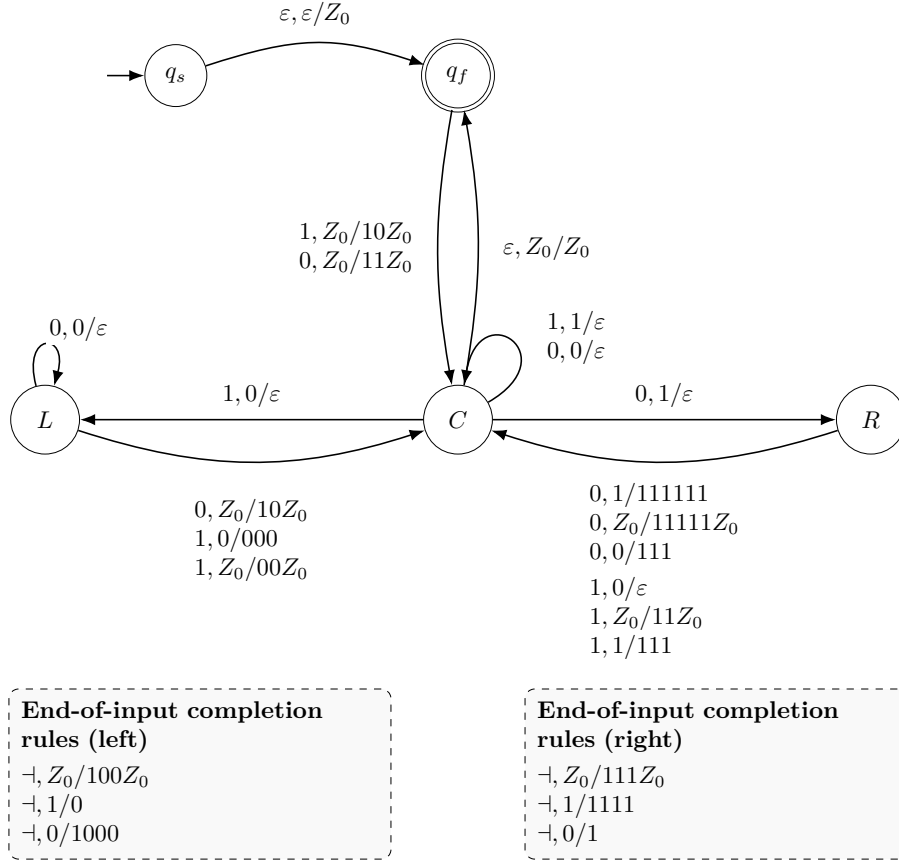


FIGURE 1. The completion-aware 2:1 deterministic push-down automaton that motivates the present paper. Solid arcs process binary input; the dashed boxes contain only right-endmarker completion rules. The machine supplies the concrete operational origin of the residual and shortest-completion viewpoints, but it is not a hypothesis of the periodic-system theorems.

The minimum number of additional symbols required to reach residual 0 is therefore a directed shortest-path distance, while the complete family of minimum completions supplies optimal first actions, shortest prefixes, alternative geodesics, and commuting local choices.

This is the abstraction step used here. Residual values are replaced by coordinates in $\mathbb{Z}^n$, finite control by a finite phase set, and completion operations by periodic partial actions carrying arithmetic displacements and

positive costs. The discovery-to-method route is

$ \begin{aligned} &2:1 \text{ completion-aware DPDA} \longrightarrow p:q \text{ residual completion} \\ &\longrightarrow \text{shortest-completion geometry} \longrightarrow \text{finite-phase periodic systems} \\ &\longrightarrow \text{cubical geodesic compilation} \end{aligned} $

Remark 1.1 (Role of the pushdown model). The 2:1 pushdown construction and its coprime $p:q$ generalization provide the concrete motivation and discovery origin of the residual, shortest-completion, and compatibility viewpoints. They are not logical prerequisites for the compiler or separation theorems developed below. Once the finite-phase periodic model is introduced, all results are proved directly in terms of arithmetic translations, shortest-path optimality, structural commutation, and optimal-compatibility complexes.

1.2. Compiler and structure theory. Cubical geodesic compilation has two layers. The first compiles metric and branching information. Finite arithmetic data represent exact shortest distances and optimal successor relations, and therefore support all optimal first moves, arbitrary shortest-prefix tests, canonical shortest routes, and an implicit exact representation of every shortest path.

The second layer applies when the system is equipped with periodic partial actions. It compiles a simplicial object

$$\mathcal{H}(v),$$

the *local optimal-compatibility complex*, whose faces represent structurally commuting action sets whose every ordering can occur as a shortest prefix.

For $h \geq 0$, write

$$\mathcal{H}_{\leq h}(v) = \{J \in \mathcal{H}(v) : |J| \leq h\}$$

and define

$$\Gamma_h(v) = (d(v), \mathcal{H}_{\leq h}(v)).$$

We prove that the hierarchy does not universally collapse at any prescribed order. For every $k \geq 1$, there exist adjacent equal-distance states r, r' such that

$$\Gamma_{k-1}(r) = \Gamma_{k-1}(r'),$$

while

$$\Gamma_k(r) \neq \Gamma_k(r').$$

More precisely,

$$\mathcal{H}_{\leq k}(r) = \mathcal{H}_{\leq k}(r') \cup \{J_\star\}$$

for one distinguished k -face $J_\star$. Independent products make these discrepancies independent and amplify them exponentially.

The resulting hierarchy separates

$ \begin{aligned} &\text{shortest-path value} < \text{optimal branching} \\ &< \text{higher-order optimal compatibility} \end{aligned} $
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1.3. Contributions. The principal contributions are as follows.

- **Cubical geodesic compilation.** We give an effective finite arithmetic compiler for positive-cost finite-phase periodic action systems. At target-reachable states it recovers exact distance, all optimal first actions, arbitrary shortest-prefix membership, the complete local optimal-compatibility complex, canonical routes, and all shortest paths implicitly.
- **Prescribed-order non-factorization.** For every $k \geq 1$, we construct a completion-total one-phase unit-cost translation system and adjacent equal-distance states whose compatibility data agree through order $k - 1$, while their truncations through order k differ by exactly one k -face.
- **No fixed lower-order reconstruction.** For every fixed h , the data $(d(v), \mathcal{H}_{\leq h}(v))$ do not universally determine $\mathcal{H}(v)$.
- **Boolean amplification.** For every $k, m \geq 1$, an m -fold product produces 2^m states with one common order- $(k - 1)$ signature and 2^m distinct order- k signatures.
- **Compiler-sensitive complexity.** We distinguish potentially expensive offline arithmetic compilation from online shortest-path queries and give query-, path-length-, and output-sensitive bounds relative to the compiled representation.

The arithmetic shortest-path backend is established in Theorem 2.5. The principal compiler theorem is Theorem 3.7. Prescribed-order separation is proved in Theorem 4.8, its reconstruction consequence appears in Theorem 4.9, and the product amplification theorem is Theorem 5.5.

1.4. Organization. Section 2 develops the arithmetic shortest-path backend for finite-phase periodic systems and establishes the finite geodesic compiler. Section 3 introduces structural cubes and local optimal-compatibility complexes, proves the cubical Bellman correspondence, and gives the finite cubical geodesic compiler together with its affine-semigroup interpretation. Section 4 constructs the binary-deficit family and proves prescribed-order adjacent-state non-factorization. Section 5 establishes the product-join theorem and Boolean amplification. Section 6 positions these results relative to periodic shortest paths, shortest-path preprocessing, cubical reconfiguration theory, Presburger arithmetic, and squarefree divisor complexes. Section 7 records compiler-relative online complexity bounds. Section 8 summarizes the resulting compilation and structure theory.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. We write n for the lattice dimension, s for the number of action labels, m for the number of independent product factors, and $d(v)$ exclusively for shortest-path distance. Distances are defined only at target-reachable states unless explicitly stated otherwise.

2. PERIODIC SHORTEST-PATH SYSTEMS

2.1. Finite-phase periodic graphs. Let

$$\mathcal{B} = (\Phi, E)$$

be a finite directed graph. Every edge

$$e : \phi \rightarrow \psi$$

carries an integral displacement

$$g_e \in \mathbb{Z}^n$$

and a positive rational cost

$$c_e \in \mathbb{Q}_{>0}.$$

Its periodic lift has vertex set

$$\mathbb{Z}^n \times \Phi$$

and edges

$$(z, \phi) \xrightarrow{e} (z + g_e, \psi).$$

Fix a target phase $\phi_\star \in \Phi$ and target $t = (0, \phi_\star)$. Multiplying all costs by a common denominator preserves all shortest paths and scales all distances uniformly. Hence we assume

$$c_e \in \mathbb{N}_{>0}.$$

A state $v = (z, \phi)$ is *target-reachable* if there exists a path from v to t . For target-reachable states, let

$$d(v) = d_\phi(z)$$

be the minimum path cost to t . When $v = (z, \phi)$, we write $\text{Reach}(v)$ as shorthand for $\text{Reach}_\phi(z)$. The same convention is used for translated successor states.

Minimum-cost path problems in periodic graphs with integral gains and rational costs are classical; see Höfting and Wanke [1]. The role of the present section is to extract a finite arithmetic representation used by the compatibility compiler.

2.2. Gain-cost relations. Let

$$L_{\phi, \phi_\star} \subseteq E^*$$

be the regular language of base-graph walks from ϕ to $\phi_\star$. For $w = e_1 \cdots e_q$, define

$$G(w) = \sum_{j=1}^q g_{e_j}, \quad C(w) = \sum_{j=1}^q c_{e_j}.$$

Set

$$R_\phi = \{(G(w), C(w)) : w \in L_{\phi, \phi_\star}\}.$$

Proposition 2.1 (Semilinear path relation). *For every phase ϕ , the relation*

$$R_\phi \subseteq \mathbb{Z}^n \times \mathbb{N}$$

is effectively Presburger-definable.

Proof. The Parikh image of a regular language is effectively semilinear [2, 3]. Total gain and total cost are integral linear functions of the edge-count vector. Semilinear sets are closed under integral linear images and coincide with Presburger-definable sets [4].

We use the standard extension of semilinearity to subsets of $\mathbb{Z}^n \times \mathbb{N}$. Equivalently, signed coordinates may be represented by nonnegative coordinate pairs and projected by the integral map $x = x^+ - x^-$; this does not change Presburger definability. $\square$

Define

$$\text{Reach}_\phi(z) \iff \exists N \in \mathbb{N} \, R_\phi(-z, N).$$

Proposition 2.2 (Arithmetic distance predicate). *There is an effectively constructible Presburger predicate*

$$\text{Dist}_\phi(z, N)$$

such that

$$\text{Dist}_\phi(z, N) \iff N = d_\phi(z)$$

for every target-reachable (z, ϕ) .

Proof. Define $\text{Dist}_\phi(z, N)$ by

$$R_\phi(-z, N) \wedge \neg \exists N' (0 \leq N' < N \wedge R_\phi(-z, N')).$$

For a target-reachable state, the feasible costs form a nonempty subset of $\mathbb{N}$, hence possess a least element. $\square$

The finite family

$$\mathfrak{C}_{\text{dist}} = \{\text{Reach}_\phi, \text{Dist}_\phi : \phi \in \Phi\}$$

will be called the *arithmetic distance compiler*. “Finite” refers to the description; the coordinate $z \in \mathbb{Z}^n$ remains unbounded.

2.3. Bellman guards. For a lifted edge $v \xrightarrow{e} T_e v$, define $e \in \text{First}(v)$ when e is the first edge of some shortest path from v to t .

Lemma 2.3 (Optimal-edge criterion). *For target-reachable v ,*

$$e \in \text{First}(v) \iff \text{Reach}(T_e v) \text{ and } c_e + d(T_e v) = d(v).$$

Proof. If a shortest path begins with e , its remaining suffix must be shortest from $T_e v$. Conversely, if the equality holds, append any shortest suffix from $T_e v$. $\square$

Lemma 2.4 (Prefix criterion). *Let $w = e_1 \cdots e_q$ be executable from a target-reachable state v . Then w is a prefix of some shortest path from v if and only if*

$$\text{Reach}(T_w v) \text{ and } C(w) + d(T_w v) = d(v).$$

Proof. If w occurs on a shortest path, replacing its suffix by a shorter one would shorten the full path, so the suffix is shortest. Conversely, append a shortest suffix from $T_w v$. The displayed equality makes the resulting total cost equal to $d(v)$. $\square$

2.4. Optimal unfolding. For a target-reachable state v , let $\text{OptUnf}(v)$ be the directed subgraph reachable from v using only edges satisfying the Bellman guard of Theorem 2.3.

Theorem 2.5 (Finite geodesic compiler). *For every fixed positive-cost finite-phase periodic graph, one can effectively construct a finite Presburger-guarded translation program such that, from every target-reachable state v ,*

- (i) *it recovers $d(v)$;*
- (ii) *it recovers every optimal first edge;*
- (iii) *it decides shortest-prefix membership for every finite executable word;*
- and*
- (iv) *its optimal unfolding is exactly the union of all shortest v -to- t paths.*

Proof. Items (i)–(iii) follow from Theorems 2.2 to 2.4. Suppose

$$v = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_q$$

is a path in the optimal unfolding. Each edge satisfies

$$d(v_{j-1}) = c_j + d(v_j),$$

hence

$$d(v_q) = d(v) - \sum_{j=1}^q c_j.$$

Every guarded successor is target-reachable, so a shortest suffix from v_q exists. Appending it yields a shortest path containing the given prefix. Conversely, every edge of a shortest path satisfies the Bellman equality by optimal substructure. $\square$

Corollary 2.6 (Canonical and complete generation). *Fix a total order on outgoing base edges. At every non-target target-reachable state, at least one optimal first edge exists. Choosing the least one gives a deterministic canonical shortest route.*

Moreover, because every cost is positive,

$$v \rightarrow v' \text{ optimal} \implies d(v') < d(v).$$

Hence d is a ranking function on the optimal unfolding. Recursive branching over all optimal first edges generates all and only shortest paths and terminates on every branch.

3. CUBICAL GEODESIC COMPILATION

3.1. Periodic action systems. Let

$$I = \{1, \dots, s\}$$

be a finite action alphabet. For every $i \in I$, let

$$\tau_i : \Phi \rightharpoonup \Phi$$

be a partial phase transition. Whenever $\tau_i(\phi)$ is defined, associate

$$g_i(\phi) \in \mathbb{Z}^n, \quad c_i(\phi) \in \mathbb{Q}_{>0}.$$

The lifted action is

$$T_i(z, \phi) = (z + g_i(\phi), \tau_i(\phi)).$$

For an executable action word $w = i_1 \cdots i_q$, T_w denotes the corresponding composition of lifted partial actions. If J is a structural cube, $T_J v$ denotes the common endpoint obtained from any permutation of the actions in J .

3.2. Structural cubes.

Definition 3.1 (Structural cube). Let $J \subseteq I$. We call J a *structural cube at phase ϕ* if, for every $K \subseteq J$, every permutation of the actions in K

- (i) is executable from ϕ ;
- (ii) reaches the same final phase;
- (iii) produces the same total displacement;
- (iv) has the same total cost;

and, additionally,

$$K \neq L \implies (\tau_K(\phi), g_K(\phi)) \neq (\tau_L(\phi), g_L(\phi)).$$

Write $\tau_K(\phi)$, $g_K(\phi)$, and $c_K(\phi)$ for the common data.

The final injectivity condition makes the $2^{|J|}$ subset vertices distinct, so the terminology “cube” is nondegenerate. Let

$$\mathcal{S}_\phi = \{J \subseteq I : J \text{ is a structural cube at } \phi\}.$$

Lemma 3.2 (Structural simpliciality). *For every ϕ , $\mathcal{S}_\phi$ is a simplicial complex.*

Proof. Every defining condition is inherited by subsets, including injectivity of the subset-endpoint map. $\square$

Because Φ and I are finite, each $\mathcal{S}_\phi$ is finite and can be computed offline from the finite action table. Cubical state complexes generated by independent local moves are classical in reconfiguration theory; see Ghrist and Peterson [5]. Our additional object is the subcomplex selected by shortest-path optimality.

3.3. Optimal compatibility.

Definition 3.3 (Optimal-compatibility complex). For a target-reachable state $v = (z, \phi)$ and $J \in \mathcal{S}_\phi$, define

$$J \in \mathcal{H}(v)$$

if

$$\text{Reach}(T_J v)$$

and

$$\boxed{c_J(\phi) + d(T_J v) = d(v)}.$$

For $h \geq 0$, define

$$\mathcal{H}_{\leq h}(v) = \{J \in \mathcal{H}(v) : |J| \leq h\}.$$

The empty set belongs to both $\mathcal{S}_\phi$ and $\mathcal{H}(v)$, with $T_\emptyset v = v$ and $c_\emptyset = 0$. Hence $\mathcal{H}_{\leq 0}(v) = \{\emptyset\}$.

Proposition 3.4 (Cubical Bellman equivalence). *Let $J \in \mathcal{S}_\phi$. For target-reachable $v = (z, \phi)$, the following are equivalent:*

- (i) $J \in \mathcal{H}(v)$;
- (ii) one permutation of the actions in J is a shortest prefix from v ;
- (iii) every permutation of the actions in J is a shortest prefix from v .

Proof. Every permutation of a structural cube has the same cost and endpoint. Hence the prefix criterion of Theorem 2.4 either holds for all permutations or for none. $\square$

Proposition 3.5 (Optimal simpliciality). *For every target-reachable v , $\mathcal{H}(v)$ is a simplicial subcomplex of $\mathcal{S}_\phi$.*

Proof. Let $J \in \mathcal{H}(v)$ and $K \subseteq J$. Order the actions of J so that the actions of K occur first. By Theorem 3.4 this permutation is a shortest prefix, and hence its initial K -segment is a shortest prefix. Therefore $K \in \mathcal{H}(v)$. $\square$

3.4. Arithmetic face guards. For each defined action i at phase ϕ , let $\text{Opt}_{\phi,i}(z)$ denote the Presburger Bellman guard supplied by Theorem 2.3. For $J \in \mathcal{S}_\phi$, define $\text{Face}_{\phi,J}(z)$ by

$$\exists N, N' \left[\text{Dist}_\phi(z, N) \wedge \text{Dist}_{\tau_J(\phi)}(z + g_J(\phi), N') \wedge N = c_J(\phi) + N' \right].$$

Thus

$$J \in \mathcal{H}((z, \phi)) \iff \text{Face}_{\phi,J}(z).$$

Definition 3.6 (Cubical geodesic compiler). Let

$$\mathfrak{C}_{\text{cube}} = \left(\{\text{Reach}_\phi, \text{Dist}_\phi\}, \{\text{Opt}_{\phi,i}\}, \{\text{Face}_{\phi,J}\} \right)$$

range over all phases, defined actions, and structural faces.

Theorem 3.7 (Finite cubical geodesic compilation). *For every fixed positive-cost finite-phase periodic action system, the finite arithmetic compiler $\mathfrak{C}_{\text{cube}}$ is effectively constructible and determines, at every target-reachable state,*

- *exact shortest distance;*
- *all optimal first actions;*
- *shortest-prefix membership for arbitrary finite executable words;*
- *the complete local optimal-compatibility complex;*
- *a canonical shortest route under fixed tie breaking; and*
- *an implicit exact representation of all shortest paths.*

Proof. The metric, first-action, prefix, canonical, and all-geodesic claims follow from Theorems 2.5 and 2.6. For each phase, the structural complex is finite. Membership of every structural face in $\mathcal{H}(v)$ is exactly one arithmetic face guard. Evaluating all such guards therefore recovers the complete local complex. $\square$

3.5. Signature regions. For a subcomplex $K \subseteq \mathcal{S}_\phi$, define

$$\Omega_{\phi,K} = \{z : \text{Reach}_\phi(z) \text{ and } \mathcal{H}((z, \phi)) = K\}.$$

Every such set is Presburger-definable, and

$$\{z : \text{Reach}_\phi(z)\} = \bigsqcup_K \Omega_{\phi,K}.$$

Thus the target-reachable region admits a finite arithmetic partition by local optimal-compatibility type.

3.6. Affine-semigroup interpretation. In the one-phase unit-cost translation case, write $\mathcal{S}$ for the unique phase structural complex. The compatibility complex admits an exact divisor-complex interpretation. Let the distinct action translations be

$$a_1, \dots, a_s \in \mathbb{Z}^n$$

and let

$$D = d(r).$$

Define

$$b_i = (-a_i, 1) \in \mathbb{Z}^{n+1}, \quad \beta_r = (r, D),$$

and the semigroup

$$\mathbf{M} = \langle b_1, \dots, b_s \rangle_{\mathbb{N}}.$$

Define the labeled squarefree divisor complex

$$\Delta_{\beta_r} = \left\{ J \subseteq [s] : \beta_r - \sum_{i \in J} b_i \in \mathbf{M} \right\}.$$

Proposition 3.8 (Homogenized semigroup correspondence). *For every $J \subseteq [s]$,*

$$J \in \Delta_{\beta_r}$$

if and only if

$$|J| + d\left(r + \sum_{i \in J} a_i\right) = d(r).$$

Consequently,

$$\boxed{\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}.$$

If every action subset is structurally nondegenerate, then $\mathcal{H}(r) = \Delta_{\beta_r}$.

Proof. Indeed,

$$\beta_r - \sum_{i \in J} b_i = \left(r + \sum_{i \in J} a_i, D - |J| \right).$$

Membership in $\mathbf{M}$ is therefore equivalent to the existence of a suffix consisting of exactly $D - |J|$ unit-cost actions that completes the state

$$r + \sum_{i \in J} a_i$$

to the target.

If the same state admitted a suffix of length strictly smaller than $D - |J|$, then preceding it by the J -prefix would give a completion of r of total length strictly smaller than

$$|J| + (D - |J|) = D = d(r),$$

a contradiction. Hence

$$d\left(r + \sum_{i \in J} a_i\right) = D - |J|,$$

which is equivalent to

$$|J| + d\left(r + \sum_{i \in J} a_i\right) = d(r).$$

Intersecting with the structural complex gives $\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}$. $\square$

Squarefree divisor complexes are classical in affine-semigroup theory [6]; parametric Presburger methods can distinguish full divisor-complex types in affine-semigroup families [7]. We therefore do not claim novelty for divisor complexes or arithmetic definability in isolation.

4. PRESCRIBED-ORDER NON-FACTORIZATION

We now show that the higher-order compatibility compiled in Section 3 cannot universally be reconstructed from any fixed lower-order truncation.

Fix $k \geq 1$ and define

$$M = 2^{k+1}, \quad \delta = 2^k - 1, \quad N = 2k.$$

Consider the one-phase unit-cost translation actions

$$a_0 = -M,$$

$$a_i = -M + 2^{i-1}, \quad 1 \leq i \leq k,$$

and

$$a_+ = 1.$$

Let

$$A_- = \{a_0, a_1, \dots, a_k\}.$$

All action maps are translations, so they commute globally.

4.1. Completion and nondegeneracy.

Lemma 4.1 (Completion-totality). *Every integer state admits a finite path to 0.*

Proof. If $x \leq 0$, repeatedly apply $a_+ = 1$. If $x > 0$, apply $a_0 = -M$ until the state becomes nonpositive and then apply a_+ . $\square$

Lemma 4.2 (Nondegeneracy of the negative family). *Distinct subsets of A_- have distinct total translations.*

Proof. For $K \subseteq A_-$, define

$$B(K) = \sum_{\substack{1 \leq i \leq k \\ a_i \in K}} 2^{i-1}.$$

Then

$$\sum_{a \in K} a = -|K|M + B(K).$$

If two subsets K, L have the same total translation, then

$$(|K| - |L|)M = B(K) - B(L).$$

The right side has absolute value $< M$, hence $|K| = |L|$. Then $B(K) = B(L)$, and uniqueness of binary expansion identifies all special actions. Cardinality then determines the presence of a_0 . Thus $K = L$. $\square$

In particular,

$$J_\star = \{a_1, \dots, a_k\}$$

is a genuine nondegenerate structural k -cube.

4.2. Binary-deficit distance.

Lemma 4.3 (Binary-deficit distance lemma). *If*

$$0 \leq D < 2^k$$

and

$$\ell \geq k,$$

then

$$\boxed{d(\ell M - D) = \ell.}$$

Proof. Write

$$D = \sum_{i \in S} 2^{i-1}.$$

Apply a_i once for every $i \in S$ and apply a_0 another $\ell - |S|$ times. The total translation is $-\ell M + D$, so $d(\ell M - D) \leq \ell$.

On the other hand, $D < 2^k < M$, so

$$\ell M - D > (\ell - 1)M.$$

One action decreases the state by at most M , so $\ell - 1$ actions cannot suffice. $\square$

Action labels used in a cubical prefix may be reused later in the completion suffix; this is ordinary path semantics.

4.3. Adjacent equal-distance states. Define

$$r = NM - \delta$$

and

$$r' = r + 1 = NM - (\delta - 1).$$

Then

$$r \xrightarrow{a_+} r'.$$

Proposition 4.4 (Equal distance).

$$\boxed{d(r) = d(r') = N.}$$

Proof. Apply Theorem 4.3 with deficits δ and $\delta - 1$. $\square$

For a negative action set J , define

$$B(J) = \sum_{\substack{1 \leq i \leq k \\ a_i \in J}} 2^{i-1}.$$

Proposition 4.5 (Exact common lower-order truncation). *For every action set J satisfying $|J| < k$,*

$$\boxed{J \in \mathcal{H}(r) \iff J \in \mathcal{H}(r') \iff J \subseteq A_-}.$$

Consequently,

$$\boxed{\mathcal{H}_{\leq k-1}(r) = \mathcal{H}_{\leq k-1}(r').}$$

Proof. Suppose first that $J \subseteq A_-$ and let $q = |J| < k$. After the prefix,

$$r_J = (N - q)M - (\delta - B(J)),$$

and

$$r'_J = (N - q)M - (\delta - 1 - B(J)).$$

Since $q < k$, not all k positive binary deficits occur, so $B(J) \leq \delta - 1$. Thus both remaining deficits belong to $[0, 2^k)$, while $N - q \geq k + 1$. Theorem 4.3 gives remaining distance $N - q$ at both states. Hence J is optimal at both.

Now suppose $a_+ \in J$. Let q be the number of negative actions in J . Then $|J| = q + 1 < k$. After the prefix from r , the residual exceeds $(N - q - 1)M$, because

$$M - \delta + B(J) + 1 > 0.$$

Yet an optimal suffix would have only $N - q - 1$ unit-cost steps available, each decreasing by at most M . This is impossible. The same argument at r'

has an extra 1 of residual and is again impossible. If such a set is structurally degenerate, it is excluded from $\mathcal{H}$ already by definition; in either case it is not an optimal face. $\square$

4.4. First discrepancy at order k .

Proposition 4.6 (Distinguished separating face). *For*

$$J_\star = \{a_1, \dots, a_k\},$$

$$J_\star \in \mathcal{H}(r), \quad J_\star \notin \mathcal{H}(r').$$

Proof. Because

$$\sum_{i=1}^k 2^{i-1} = \delta,$$

after applying $J_\star$ at r ,

$$r + \sum_{i=1}^k a_i = (N - k)M = kM.$$

Theorem 4.3 with $D = 0$ gives $d(kM) = k = N - k$. Hence $J_\star$ is optimal.

At r' , the same prefix leaves $kM + 1$. The required suffix would have k actions, which can decrease the residual by at most kM . Hence no such optimal suffix exists. $\square$

Proposition 4.7 (Single-face difference through order k).

$$\mathcal{H}_{\leq k}(r) = \mathcal{H}_{\leq k}(r') \cup \{J_\star\}.$$

Proof. The complexes agree below order k by Theorem 4.5. Any k -set containing a_+ fails the same maximum-descent test at both states.

A negative k -set other than $J_\star$ contains a_0 and omits exactly one special action a_i . Its deficit is $\delta - 2^{i-1}$. After the prefix, the two residuals are

$$kM - 2^{i-1}$$

and

$$kM - (2^{i-1} - 1),$$

respectively. Both deficits lie in $[0, 2^k)$, so Theorem 4.3 gives remaining distance k in both cases. $\square$

Theorem 4.8 (Prescribed-order adjacent-state non-factorization). *For every prescribed $k \geq 1$, there exists a completion-total one-phase unit-cost system of globally commuting translation actions and adjacent states $r \rightarrow r'$ such that*

$$d(r) = d(r'),$$

$$\mathcal{H}_{\leq k-1}(r) = \mathcal{H}_{\leq k-1}(r'),$$

and

$$\mathcal{H}_{\leq k}(r) = \mathcal{H}_{\leq k}(r') \cup \{J_\star\}.$$

Equivalently,

$$\Gamma_{k-1}(r) = \Gamma_{k-1}(r')$$

but

$$\Gamma_k(r) \neq \Gamma_k(r').$$

Thus the first compatibility discrepancy occurs exactly at the prescribed order k .

Corollary 4.9 (No fixed lower-order reconstruction). *For every fixed $h \geq 0$, the data*

$$(d(v), \mathcal{H}_{\leq h}(v))$$

do not universally determine $\mathcal{H}(v)$.

Proof. Apply Theorem 4.8 with $k = h + 1$. □

The claim concerns this exact lower-order truncation. It does not assert insufficiency of an arbitrary auxiliary state description that might retain additional information such as the raw coordinates.

4.5. Exact full complexes.

Proposition 4.10 (Full-complex description).

$$\mathcal{H}(r) = 2^{A_-}$$

and

$$\mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\}.$$

Proof. Every proper negative subset of size at most k is handled by Theorems 4.3, 4.5 and 4.7. For the full negative set,

$$r + \sum_{a \in A_-} a = (k - 1)M.$$

If $k = 1$, this is already the target. If $k \geq 2$, its distance is $k - 1$ by maximal descent and the obvious a_0 -completion. Hence $A_- \in \mathcal{H}(r)$.

At r' , the same prefix leaves $(k - 1)M + 1$, which cannot be completed in the required $k - 1$ steps. No face containing a_+ is optimal. If its prefix cost already exceeds N , this is immediate; otherwise the same maximum-descent argument used above applies. □

Accordingly,

$$\text{through order } k \text{ the complexes differ by one } k\text{-face,}$$

whereas

$$\text{the full complexes differ by two faces.}$$

This distinction is retained throughout the paper.

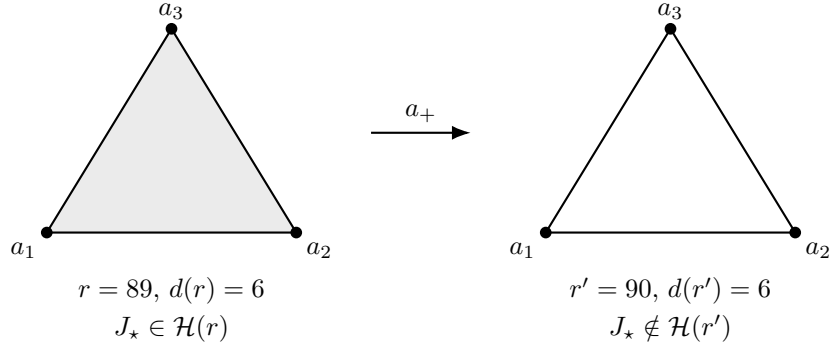


FIGURE 2. Prescribed-order separation for $k = 3$. The adjacent states $r = 89$ and $r' = 90$ have equal shortest distance and identical optimal-compatibility data through order 2. On the distinguished action set $J_\star = \{a_1, a_2, a_3\}$, the restriction of the optimal-compatibility complex changes from a filled triangle to its boundary. Thus the first discrepancy occurs exactly at order 3.

4.6. Large optimization-induced minimal nonfaces. For $k \geq 2$, on the distinguished vertex set $J_\star$,

$$\mathcal{H}(r)|_{J_\star} = \Delta^{k-1},$$

while

$$\mathcal{H}(r')|_{J_\star} = \partial\Delta^{k-1}.$$

For $k = 1$, the distinguished vertex is present at r and absent at r' . Thus shortest-path optimality creates minimal nonfaces of arbitrarily large cardinality inside a globally commuting translation system. For $k \geq 3$, the resulting restricted complex is nonflag.

The novelty claim is not the existence of nonflag complexes or simplex boundaries in isolation; such phenomena are classical in semigroup topology [6]. The relevant constraint package is adjacency, equal shortest distance, one fixed action system, agreement of all lower compatibility orders, and first discrepancy at a prescribed order.

Remark 4.11 (Computational verification). As an independent regression check, the binary-deficit family was exhaustively verified for $1 \leq k \leq 10$. The distance formula, the exact lower-order agreement, and the full-complex identities of Theorem 4.10 matched the symbolic formulas in every case. Independent product-graph checks were also performed for representative small (k, m) pairs and recovered exactly the Boolean signatures of Theorem 5.5. These checks are not premises of any theorem.

5. BOOLEAN AMPLIFICATION

We now take independent products of the separating block.

5.1. Product formulas. Consider m periodic action systems with disjoint action alphabets.

Lemma 5.1 (Product distance). *For*

$$V = (v_1, \dots, v_m),$$

$$d(V) = \sum_{t=1}^m d_t(v_t).$$

Proof. Every product path projects to a path in each factor, giving the lower bound. Interleaving shortest factor paths gives equality. $\square$

For $v_t = (z_t, \phi_t)$, write $\mathcal{S}_t(\phi_t)$ for the structural complex in factor t , and write $\mathcal{S}(V)$ for the product structural complex at V . For an action set

$$J = J_1 \sqcup \dots \sqcup J_m,$$

structural compatibility factors:

$$J \in \mathcal{S}(V) \iff J_t \in \mathcal{S}_t(\phi_t) \quad \forall t.$$

Distinct factor-subset tuples give distinct product endpoints, so nondegeneracy also factors.

Lemma 5.2 (Product-join lemma).

$$\mathcal{H}(V) = \mathcal{H}_1(v_1) * \dots * \mathcal{H}_m(v_m).$$

Proof. First, $T_J V$ is target-reachable exactly when every $T_{J_t} v_t$ is target-reachable. In that case define

$$\sigma_t = c_{J_t} + d_t(T_{J_t} v_t) - d_t(v_t) \geq 0.$$

Then

$$c_J + d(T_J V) - d(V) = \sum_{t=1}^m \sigma_t.$$

The sum vanishes exactly when every σ_t vanishes. $\square$

5.2. Product family. Take m independent copies of the Section 4 block. Write

$$r^{(0)} = r, \quad r^{(1)} = r',$$

and for $\varepsilon \in \{0, 1\}^m$ define

$$R^\varepsilon = (r^{(\varepsilon_1)}, \dots, r^{(\varepsilon_m)}).$$

Let $J_\star^{(t)}$ be the distinguished k -face in factor t .

Proposition 5.3 (Constant lower-order signature). *For every $\varepsilon, \eta \in \{0, 1\}^m$,*

$$d(R^\varepsilon) = d(R^\eta) = mN$$

and

$$\Gamma_{k-1}(R^\varepsilon) = \Gamma_{k-1}(R^\eta).$$

Proof. Distance additivity gives the first assertion. Every factor component of an action set of total size at most $k - 1$ has size at most $k - 1$, where the two base states have identical compatibility. Apply Theorem 5.2. $\square$

Proposition 5.4 (Exact order- k variation). *There exists a fixed complex $\mathcal{K}_{k,m}$ such that*

$$\mathcal{H}_{\leq k}(R^\varepsilon) = \mathcal{K}_{k,m} \cup \{J_\star^{(t)} : \varepsilon_t = 0\}.$$

Proof. If an order- $\leq k$ face uses more than one factor, then every factor component has size $< k$, so its membership is determined entirely by the common lower-order base data. If all k actions lie in one factor, Theorem 4.7 says the only variable face is that factor's distinguished $J_\star$. $\square$

Theorem 5.5 (Boolean fiber amplification). *For every $k, m \geq 1$, there exists a completion-total one-phase unit-cost translation system containing 2^m states with one common order- $(k - 1)$ signature and 2^m pairwise distinct order- k signatures.*

The variable part of the signature is precisely

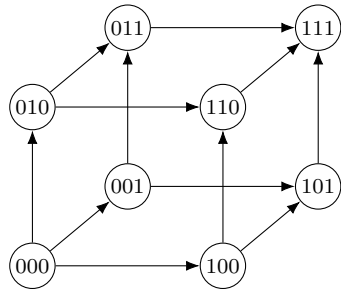
$$\{J_\star^{(t)} : \varepsilon_t = 0\},$$

so the constructed order- k signatures form a Boolean lattice of rank m .

The states themselves form a directed m -cube in the underlying state graph, because changing $\varepsilon_t : 0 \rightarrow 1$ is exactly the factor action $a_+^{(t)}$. Throughout this state cube,

$$d(R^\varepsilon) = mN$$

and $\Gamma_{k-1}(R^\varepsilon)$ remain constant, whereas Γ_k distinguishes every vertex.



$$d(R^\varepsilon) = 3N \text{ for all } \varepsilon$$

$$\Gamma_{k-1}(R^\varepsilon) \text{ is constant}$$

$$\Gamma_k \text{ distinguishes all 8 vertices}$$

$$\text{coordinate } t \text{ toggles } J_\star^{(t)}$$

FIGURE 3. Boolean amplification for three independent factors. The eight product states form a directed state cube. Exact distance and the complete order- $(k - 1)$ signature are constant throughout the cube. Moving in coordinate t toggles exactly the distinguished order- k face $J_\star^{(t)}$, producing eight distinct order- k signatures.

5.3. Exponential fibers. Each factor has $k + 2$ action labels. Thus the product has

$$s = m(k + 2)$$

actions.

Corollary 5.6 (Exponential signature fibers). *For every fixed $k \geq 1$ and every $m \geq 1$, there is a system with*

$$s = m(k + 2)$$

actions for which one order- $(k - 1)$ signature supports at least

$$\boxed{2^m = 2^{s/(k+2)}}$$

distinct order- k signatures. Thus exponentially large fibers occur along the infinite sequence of action dimensions divisible by $k + 2$.

5.4. Conditional discrimination. Suppose an auxiliary summary $C(v)$ is required to support an exact decoder

$$\Gamma_k(v) = F(\Gamma_{k-1}(v), C(v))$$

on the constructed family, with no raw state coordinate supplied separately.

Corollary 5.7 (Conditional discrimination bound).

$$\boxed{|\text{im } C| \geq 2^m}$$

and therefore

$$\boxed{\log_2 |\text{im } C| \geq m = \frac{s}{k+2}.$$

Proof. All R^ε have the same lower-order signature but pairwise distinct order- k signatures. Hence two distinct states cannot receive the same summary value. $\square$

This is a discrimination bound for exact summaries under the stated decoder model. It is not a general RAM-space lower bound.

6. RELATION TO PREVIOUS WORK

Periodic minimum-cost paths are classical. Höfting and Wanke study directed finite descriptions with integral vector gains and rational costs and ask for minimum-cost paths realizing prescribed displacements [1]. The distance backend of Section 2 should therefore be viewed as arithmetic infrastructure rather than as the principal novelty of the paper.

The semilinear and Presburger ingredients are likewise classical. Parikh images provide semilinear descriptions of path-count data [2], while Ginsburg and Spanier identify semilinear sets with Presburger-definable sets [4]. Modern work gives explicit constraint constructions for regular-language Parikh images [3].

Offline shortest-path preprocessing also has a substantial literature. Compressed Path Databases precompute shortest-path information and retrieve

optimal moves at query time rather than rerunning graph search [8, 9]. Cubical geodesic compilation shares this compile-once/query-many perspective, but its state space is periodic and infinite and its output extends to arbitrary-order joint optimal compatibility.

Cubical representations of independent actions appear prominently in reconfiguration theory. Ghrist and Peterson associate cubical state complexes with independent local moves and study both their geometry and spaces of optimal reconfiguration paths [5]. We therefore do not claim novelty for commuting moves forming cubes, or for cubical topology of optimal path spaces in isolation.

The one-phase unit-cost optimal-compatibility complex is closely tied to squarefree divisor complexes, as Theorem 3.8 makes exact. Such complexes have a long algebraic history and may realize extremely general topology [6]. Parametric Presburger arithmetic has also been used to distinguish complete squarefree-divisor-complex types in affine-semigroup families [7].

Accordingly, the new separation theorem is deliberately narrower and more constrained. It concerns one fixed globally commuting translation system, adjacent states, equal exact shortest distance, agreement at every compatibility order below a prescribed k , and a first discrepancy at exactly that order. We are not aware of a result in the above literatures establishing this prescribed-order adjacent-state separation.

7. COMPLEXITY OF THE COMPILED REPRESENTATION

The compiler has two distinct computational stages:

$$\boxed{\text{offline arithmetic compilation} \quad + \quad \text{online querying.}}$$

No polynomial-time claim is made for the general offline stage.

7.1. Query parameters. For $v = (z, \phi)$, let

$$\lambda(v) = 1 + \sum_{j=1}^n \lceil \log_2(1 + |z_j|) \rceil$$

be the coordinate encoding length.

Let $Q_{\text{reach}}(\lambda)$ denote the cost of evaluating target reachability and let $Q_{\text{dist}}(\lambda)$ denote the additional cost of recovering the exact distance once reachability has been established. We write

$$Q_{\text{met}}(\lambda) = Q_{\text{reach}}(\lambda) + Q_{\text{dist}}(\lambda)$$

for a complete metric query. This convention is necessary because $d(v)$ is defined only at target-reachable states. In particular, successor states appearing in optimal-action, prefix, and face queries are tested for reachability before their distance values are evaluated.

Let $Q_{\text{opt}}(\lambda)$ and $Q_{\text{face}}(\lambda)$ denote the costs of evaluating one compiled optimal-action guard and one face guard, respectively. Let $Q_{\text{first}}(\lambda)$ denote

the cost of recovering the complete set of optimal first actions at one target-reachable state. Let $\deg_{\max}^+$ be the maximum outgoing base degree.

Proposition 7.1 (Distance and first-action queries). *Exact distance at a target-reachable state is recovered in $Q_{\text{dist}}(\lambda(v))$. All optimal first actions are recovered in*

$$O(\deg_{\max}^+ Q_{\text{opt}}(\lambda_*)).$$

If optimal guards are evaluated through metric queries instead, then

$$O(\deg_{\max}^+ Q_{\text{met}}(\lambda_*) + Q_{\text{dist}}(\lambda(v)))$$

suffices.

7.2. Prefix queries. Let w be an explicit executable word of length q . A single scan computes its endpoint phase, accumulated gain, and accumulated cost. For a fixed system,

$$\lambda(T_w v) = O(\lambda(v) + n \log(1 + q)).$$

Proposition 7.2 (Prefix-query complexity). *Shortest-prefix membership is decidable in*

$$O(q \text{ poly}(\lambda_w) + Q_{\text{met}}(\lambda_w) + Q_{\text{dist}}(\lambda(v))).$$

The linear dependence on q is unavoidable up to arithmetic costs because the explicit word must be read.

7.3. Face queries. A fixed structural face has precomputed phase, displacement, and cost data.

Proposition 7.3 (Face-query complexity). *A specified structural face can be tested in $Q_{\text{face}}(\lambda_*)$. Using only reachability and distance recovery gives*

$$O(Q_{\text{dist}}(\lambda(v)) + Q_{\text{met}}(\lambda(T_J v))).$$

Let

$$F_h(\phi) = |\{J \in \mathcal{S}_\phi : |J| \leq h\}|.$$

Then

$$F_h(\phi) \leq \sum_{j=0}^h \binom{s}{j}.$$

Thus the complete explicit truncation $\mathcal{H}_{\leq h}(v)$ can be reported in

$$O(F_h(\phi) Q_{\text{face}}(\lambda_*) + |\mathcal{H}_{\leq h}(v)|).$$

Explicit recovery of the full complex can require exponential output, since even a full simplex has 2^s faces.

7.4. Canonical routes. Let

$$c_{\min} = \min_e c_e > 0.$$

Every optimal edge decreases distance by at least $c_{\min}$. Therefore any shortest route has at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor$$

edges. If $\ell_{\text{can}}(v)$ is the canonical route length and Q_{canon} is the cost of selecting the canonical enabled optimal action, then route generation takes

$$O(\ell_{\text{can}}(v)Q_{\text{canon}}(\lambda_{\text{path}})).$$

This is path-length-sensitive. With binary-encoded coordinates, an explicitly printed shortest path may itself be exponentially long in the input bit length.

7.5. Enumeration of all shortest paths. Let $\text{Geo}(v)$ be the finite set of shortest paths from v , and define

$$\text{OUT}(v) = 1 + \sum_{\gamma \in \text{Geo}(v)} (|\gamma| + 1).$$

The shortest-prefix tree has at most $O(\text{OUT}(v))$ nodes.

Theorem 7.4 (Output-sensitive geodesic enumeration). *All shortest paths from v can be enumerated in*

$$O(\text{OUT}(v)[Q_{\text{first}}(\lambda_{\text{path}}) + 1]).$$

The recursion depth is at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor.$$

We describe this as output-sensitive and path-length-sensitive rather than polynomial-delay enumeration, because a single shortest path may already be exponentially long in the binary coordinate encoding.

7.6. Normalized arithmetic compilers. A stronger online statement is available under an explicit normalization assumption. Suppose offline preprocessing produces

- (i) a finite decision list of quantifier-free Presburger regions;
- (ii) an explicit affine distance output on each region; and
- (iii) quantifier-free optimal-action and face guards.

Let $\|\mathfrak{C}\|$ be the encoded compiler size and $L_{\mathfrak{C}}$ the maximum coefficient bit length.

Corollary 7.5 (Online bit complexity after normalization). *Under this explicit normalization assumption, DIST, FIRST, and FACE are computable in*

$$\text{poly}(\|\mathfrak{C}\|, L_{\mathfrak{C}}, \lambda(v)).$$

For a fixed compiled system, these queries are therefore polynomial in the queried coordinate bit length.

This is a conditional normalized-compiler result; the paper does not claim that the required normalization is polynomial-time constructible in general. General Presburger elimination can incur substantial blow-up. Recent work obtains singly exponential elimination for an existential block, but this does not imply polynomial complexity for the complete compilation pipeline [10].

8. CONCLUSION

We developed *cubical geodesic compilation*, a finite arithmetic method for positive-cost finite-phase periodic shortest-path systems. The method arose from the shortest-completion geometry of 2:1 deterministic pushdown normalization and its coprime $p:q$ generalization, but the resulting compiler and structure theorems are formulated and proved entirely at the periodic-system level.

The compiler records more than metric value. At every target-reachable state it recovers exact shortest distance, all optimal first moves, arbitrary shortest-prefix membership, canonical shortest routes, an implicit exact representation of all shortest paths, and the complete local optimal-compatibility complex.

The structure theory shows that the last of these objects contains genuinely higher-order information. For every prescribed $k \geq 1$, adjacent equal-distance states can have identical optimal-compatibility data through order $k - 1$ while their truncations through order k differ by exactly one k -face. Consequently, for every fixed h , the pair

$$(d(v), \mathcal{H}_{\leq h}(v))$$

does not universally determine the full local optimal-compatibility complex.

Independent products make these discrepancies simultaneous and independent. For every $k, m \geq 1$, one common order- $(k - 1)$ signature supports 2^m distinct order- k signatures, with m distinguished faces toggled independently.

The resulting picture is therefore two-sided:

shortest-path geometry is finite in arithmetic description

while

optimal-compatibility information is unbounded in order across the class.

Thus finite compilation and unbounded informational order coexist within the same periodic shortest-path framework.

DECLARATIONS

Author contributions. Alp Eren Bütün: Conceptualization, Methodology, Formal analysis, Investigation, Computational verification, Visualization, Writing – original draft, Writing – review & editing. The sole author developed the mathematical framework, formulated and proved the results, carried out the computational regression checks, prepared the figures, and wrote and revised the manuscript.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest. The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. The paper is theoretical and does not use an external dataset. The finite computational instances used for regression checks are generated directly from the explicit constructions stated in the manuscript. No empirical dataset is required to reproduce or verify the mathematical results.

Code availability. No software package is required to establish the results of the paper. The computational checks described in the manuscript were used as independent regression support and are not premises of any theorem.

Computational verification. Selected formulas and finite instances were checked computationally as an independent regression procedure. These checks include the binary-deficit construction, the exact local optimal-compatibility complexes for finite parameter ranges, and representative product constructions. All theorems are proved mathematically in the manuscript and do not rely on computational verification.

Ethics approval and consent. Not applicable. The work is purely mathematical and does not involve human participants, animals, clinical data, or personal data.

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